Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | Paper -2(Adv-2024-P2-Model) | Date: 07-09-2025

Time: 02.00Pm to 05.00Pm | GTA-01 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**B**	2	**B**	3	**C**	4	**A**	5	**BCD**	6	**AD**
7	**ACD**	8	**1**	9	**30**	10	**11**	11	**10**	12	**2000**
13	**125**	14	**300**	15	**1200**	16	**0.33**	17	**0.75**		

PHYSICS

18	**A**	19	**B**	20	**C**	21	**B**	22	**ABC**	23	**AC**
24	**AD**	25	**2**	26	**1**	27	**6**	28	**4**	29	**3**
30	**5**	31	**4**	32	**8**	33	**18**	34	**4**		

CHEMISTRY

35	**B**	36	**C**	37	**C**	38	**C**	39	**ACD**	40	**AC**
41	**ABC**	42	**2**	43	**2**	44	**300**	45	**19**	46	**3**
47	**8**	48	**25**	49	**9**	50	**29.6 - 30.5**	51	**66.6**		

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SOLUTIONS

MATHEMATICS

1. $f(x)=\begin{cases} x-1; & x=\text{Even} \\ 2x; & x=\text{odd} \end{cases}$

$f(f(f(a)))=1$

C-1: If $a=\text{even}$

$f(a)=a-1=\text{odd}$

$f(f(a))=2(a-1)=\text{even}$

$f(f(f(a)))=2a-3=1\Rightarrow a=2$

C-2: If $a=\text{odd}$

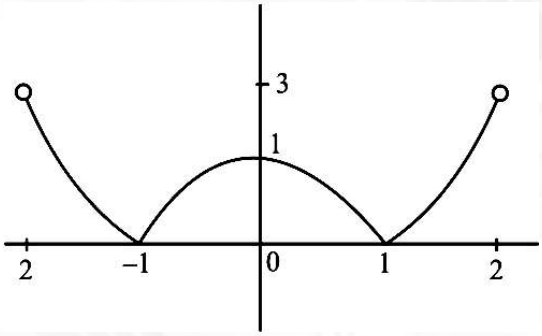
$f(a)=2a=\text{even}$

$f(f(a))=2a-1=\text{odd}$

$f(f(f(a)))=4a-2=1$ (Not possible)

Hence $a=2$

$\therefore g(x)=|x^2-1|;x\in(-2,2)$

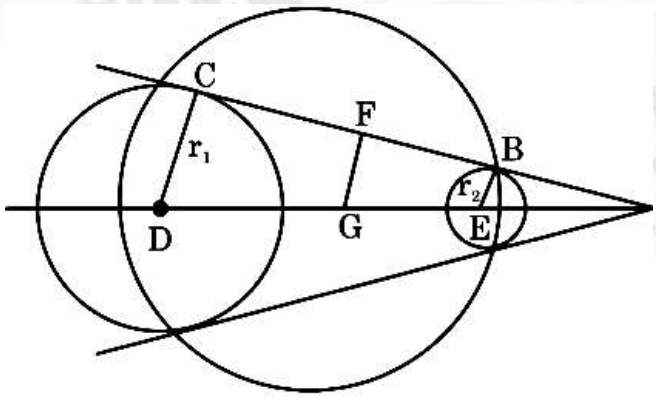


$\Rightarrow$ continuous everywhere not differentiable at $x=-1,1$

$\therefore m=0,n=2$

Hence $m+n=2$

2.



$FG=\dfrac{r_1+r_2}{2}$

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$$GB=\sqrt{(FG)^2+(FB)^2}$$

$$(GB)^2=\frac{(r_1+r_2)^2}{4}+\frac{d^2-(r_1-r_2)^2}{4}=\frac{d^2+4r_1r_2}{4}$$

$$R_1^2=\frac{(C_1C_2)^2+4r_1r_1}{4}=40$$

$$R_2^2=\frac{(C_1C_2)^2-4r_1r_1}{4}=10$$

3. Given equation simplifies to

$$\left(4\sin^2 x-2\sin^2 z\right)^2+4(2\sin x.\sin z-\sin y)^2+2\left(\sin^2 y-1\right)^2=0$$

$$\therefore \sin^2 y=1, 2\sin x.\sin z=\sin y, 2\sin^2 x=\sin^2 z$$

Cases	No. of triplets
$\sin y=1,$ sin $\sin x=\left(\frac{1}{8}\right)^4, \sin z=\left(\frac{1}{2}\right)^4$	4
$\sin y=1,$ $\sin x=-\left(\frac{1}{8}\right)^4, \sin z=-\left(\frac{1}{2}\right)^4$	4
$\sin y=-1,$ $\sin x=\left(\frac{1}{8}\right)^4, \sin z=-\left(\frac{1}{2}\right)^4$	4
$\sin y=-1,$ $\sin x=-\left(\frac{1}{8}\right)^4, \sin z=+\left(\frac{1}{2}\right)^4$	4

4. $h(x)=(x-\alpha_1)(x-\alpha_2).....(x-\alpha_5)$ take log both side and differentiate

$$\frac{h'(x)}{h(x)}=\frac{1}{x-\alpha_1}+\frac{1}{x-\alpha_2}+.....+\frac{1}{x-\alpha_5}$$

Again diff.

$$\frac{h.h''-(h')^2}{h^2}<0\forall x\in R$$

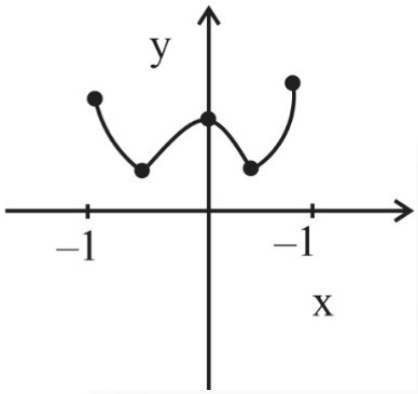
No real roots

5. Put $z=e^{i\theta}$

$$|1-z|=2\left|\sin\frac{\theta}{2}\right| \text{ and } \left|1+z^2\right|=\sqrt{(1+\cos 2\theta)^2+\sin^2 2\theta}=2\left|1-2\sin^2\frac{\theta}{2}\right|$$

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Hence, $y=|x|+\left|1-2x^2\right|\in\left[\frac{\sqrt{2}}{2},2\right];x\in[-1,1]$ where $x=\sin\frac{\theta}{2}$



Hence, $|1-z|+\left|1+z^2\right|\in|\sqrt{2},4|$

6. $\lim\limits_{x\to0}\left(\frac{1}{x^{2024}}\int\limits_{x^{2025}}^{x^{2024}}\left(\frac{(1+t)^{\frac{1}{2023}}-1}{t}\right)dt\right)=\frac{1}{P}$

$$\frac{\frac{\left(\left(1+x^{2024}\right)^{\frac{1}{2023}}-1\right)2024x^{-2023}}{x^{2024}}-\frac{\left(\left(1+x^{2025}\right)^{\frac{1}{2023}}-1\right)2025x^{2024}}{x^{2025}}}{2024x^{2023}}$$

$$\frac{2024\left(\left(1+x^{2024}\right)^{\frac{1}{2023}}-1\right)}{2024x^{2024}}-\frac{\left(\left(1+x^{2025}\right)^{\frac{1}{2023}}-1\right)2025}{2024x^{2024}}=\frac{1}{2023}=\frac{1}{P}$$

(A) $(2023)^{2023}=(2030-7)^{2023}=(35K-7)^{2023}$

$={}^{2023}C_0(35K)^{2023}(-7)^0+{}^{2023}C_1(35K)^{2022}(-7)+....+$

$.....+{}^{2023}C_{2023}(-7)^{2023}=35N-7^{2023}.$

Now, $-7^{2023}=-7\times7^{2022}=-7\left(7^2\right)^{1011}=-7(50-1)^{1011}$

$=-7\left({}^{1011}C_0\,50^{1011}-{}^{1011}C_1(50)^{1010}+.....-{}^{1011}C_{1011}\right)=-7(5\lambda-1)=-35\lambda+7$

$\therefore$ When $(2023)^{2023}$ is divided by 35, remainder is 7.

(B) $\ell n\,x=2023x$ has exactly NO real solutions

(C) $(2020)^x+(2021)^x+(2022)^x=(2023)^x$

Let $g(x)=\left(\frac{2020}{2023}\right)^x+\left(\frac{2021}{2023}\right)^x+\left(\frac{2022}{2023}\right)^x-1$

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$\because g(x)$ is a decreasing function & $\lim_{x\to\infty} g(n) = -1$ $\Rightarrow$ It has only one root

(D) Unit digit of $(2023)^{2023}$ is '7.'

7.

Interchange column twice ↑ Transposing & interchange row twice ↑

$$f(x) = \begin{vmatrix} 5 & 7 & 3 \\ 2 & 9g^2x & 2x^2 \\ x & 3g(x)-x^2 & 1 \end{vmatrix} - \begin{vmatrix} 4 & 5 & 2 \\ 2 & 9g^2x & 2x^2 \\ x & 3g(x)-x^2 & 1 \end{vmatrix} + \begin{vmatrix} 0 & -2 & 1 \\ 2 & 9g^2(x) & 2x^2 \\ x & 3g(x)-x^2 & 1 \end{vmatrix}$$

$$f(x) = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 9g^2(x) & 2x^2 \\ x & 3g(x)-x^2 & 1 \end{vmatrix} = 9g^2(x) - 6x^2g(x) + 2x^4$$

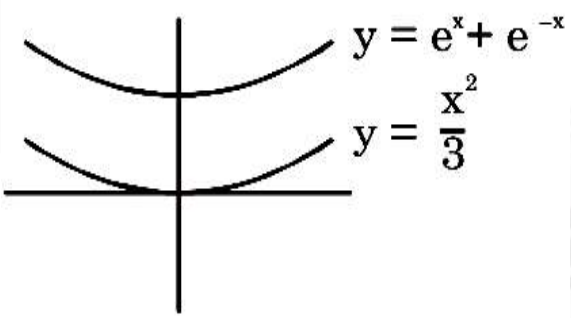
$$(i)\, m = \int_0^1 \left[\left(3g(x)-x^2\right)^2 + x^4 \right] dx \text{ for minimum}$$

$$\therefore m = \int_0^1 x^4 dx = \frac{1}{5} \text{ and } g(x) = \frac{x^2}{3}$$

$$(ii)\, \phi(x) = \int_0^x \left(\frac{t^3}{9} - e^t + e^{-t} \right) dt$$

$$\phi''(x) = \frac{x^2}{3} - e^x - e^{-x} = 0$$

$$\therefore \frac{x^3}{3} = e^x + e^{-x}$$



No solution

8. $\int_0^{f(x)} f^{-1}(t)\,dt - \int_0^x \left(\cos x - f(x)\right) dt = 0$

Differentiate w.r.t. to x

$\Rightarrow f^{-1}\left(f(x)\right).f'(x) - \left(\cos x - f(x)\right) = 0$

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$\Rightarrow xf'(x)+f(x)=\cos x$

$\Rightarrow (xf(x))'=\cos x$

$\Rightarrow xf(x)=\sin x+C$

As $f\left(\frac{\pi}{2}\right)=\frac{2}{\pi}\Rightarrow C=0$

So $f(x)=\frac{\sin x}{x}$

Now, $\int_0^{\pi/2} f(x)dx=\int_0^{\pi/2}\frac{\sin x}{x}dx$

As, $\frac{2}{\pi}<\frac{\sin x}{x}<1$

$1<\int_0^{\pi/2}\frac{\sin x}{x}dx<\frac{\pi}{2}$

So, $\left[\int_0^{\pi/2}\frac{\sin x}{x}dt\right]=1$

9. $2A+3BB^T=1,\ B^{-1}=A^T$ .....(i)

$\left(2A+3BB^T\right)^T=I$

$2A^T+3BB^T=I$ .....(ii)

$\therefore A=A^T$ also $A^T=B^{-1}$

$\therefore A=B^{-1}$ or $A^{-1}=B$ .....(iii)

$AB=BA=I=BB^{-1}$

Now $A^T=B^{-1}$

$A^{-1}B=1$

From (iii) $\therefore B=A^{-1}$

$\therefore B^T=\left(A^{-1}\right)^T=\left(A^T\right)^{-1}$

$B^T=\left(B^{-1}\right)^{-1}\therefore B=B^T$

$2A+3B^2=I\ \ \therefore 2AB+3B^3=B$

$2I+3B^3=B\ \ \therefore 3B^3=B-2I$

Let $\therefore \alpha=\left|A^{-1}-3B^3+BA\right|=\left|B-(B-2I)+I\right|=|3I|=27$

$\beta=tr\left(A^{-1}-3B^3-BA\right)=tr\left((B-(B-2I)-I)\right)$

$\beta=tr(I)=3$

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10. $f(x)=x^3+(x-1)^3-4x+\frac{5}{2}$

$f(1-x)=(1-x)^3+(-x)^3-4(1-x)+\frac{5}{2}$

$\Rightarrow f(x)+f(1-x)=1$

Let $I=\int_{\frac{1}{8}}^{\frac{7}{8}} f(f(x))dx$ .....(1)

Apply king property

$I=\int_{\frac{1}{8}}^{\frac{7}{8}} f(1-f(x))dx$ .....(2)

Adding (1) and (2)

$2I=\int_{\frac{1}{8}}^{\frac{7}{8}} f(f(x))dx+\int_{\frac{1}{8}}^{\frac{7}{8}} f(1-f(x))dx$

$2I=\int_{\frac{1}{8}}^{\frac{7}{8}} dx=\frac{3}{4}$

$\Rightarrow I=\frac{3}{8}$

M=3, n=8 m+n=11

11. The given sequence 4,7,1,8,9,7,6,3,9,2,1,3,4,7,1

$S_{12}=4+7+1+.....+1+3=60$

$S_n>10004$

$\frac{10004}{60}=166$ and $166\times12=1992$

$S_{1992}=60\times166=9960$

9960+4+7+1+8+9+7+6=10002+3=10005

No. of terms = 2000

$S_{2000}>10004$

12. Let $f(x)=(1+x)\left(1+x^2\right)\left(1+x^3\right).....\left(1+x^{2000}\right)$

$f(x)=\sum_{n=0}^{N} a_n x^n$

Our answer is $a_5+a_{10}+a_{15}+.....$

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Let $\alpha = e^{i2\pi/5}$

$f(x) = a_0 + a_1x + a_2x^2 + ... + a_5x^5 + ... + a_{10}x^{10} + ...$

$f(1) = a_0 + a_1 + a_2 + ... + a_5 + a_6 + a_7 + ...$

$f(\alpha) = a_0 + a_1\alpha + a_2\alpha^2 + ... + a_5\alpha^5 + a_6\alpha^6 + ...$

$f(\alpha^2) = a_0 + a_1\alpha^2 + a_2\alpha^4 + ...$

$f(\alpha^3) = a_0 + a_1\alpha^3 + ... + a_5\alpha^{15} + ...$

$f(\alpha^4) = a_0 + a_1\alpha^4 + ... + a_5\alpha^{20} + ...$

Add $\Rightarrow f(1) + f(\alpha) + f(\alpha^2) + f(\alpha^3) + f(\alpha^4) = 5(a_0 + a_5 + a_{10} + a_{15} + ...)$

$\Rightarrow a_0 + a_5 + a_{10} + a_{15} + ... = \dfrac{f(1) + f(\alpha) + f(\alpha^2) + f(\alpha^3) + f(\alpha^4)}{5}$

Now $f(x) = (1+x)(1+x^2)(1+x^3)(1+x^4)(1+x^5)....(1+x^{2000})$

$f(1) = 2^{2000}$

$f(\alpha) = \left[(1+\alpha)(1+\alpha^2)(1+\alpha^3)(1+\alpha^4)(1+\alpha^5)\right]^{400}$

$\because x^5 - 1 = (x-\alpha)(x-\alpha^2)(x-\alpha^3)(x-\alpha^4)(x-\alpha^5)$

$-2 = -(1+\alpha)(1+\alpha^2)(1+\alpha^3)(1+\alpha^4)(1+\alpha^5)$

$\Rightarrow f(\alpha) = \alpha^{400}$

Similarly $f(\alpha) = f(\alpha^2) = f(\alpha^3) = f(\alpha^4) = 2^{400}$

$\Rightarrow a_0 + a_5 + a_{10} + a_{15} + ... = \dfrac{2^{2000} + 4(2)^{400}}{5}$

$\Rightarrow n = \dfrac{2^{2000} + 2^{402}}{5} - 1$

$5(n+1) = 2^{402}\left(2^{1598} + 1\right)$

13. $\left[\bar{\alpha} \quad \bar{\beta} \quad \bar{\gamma}\right] = \begin{vmatrix} \bar{a}.\bar{a} & \bar{a}.\bar{b} & \bar{a}.\bar{c} \\ \bar{a}.\bar{b} & \bar{b}.\bar{b} & \bar{b}.\bar{c} \\ \bar{a}.\bar{c} & \bar{b}.\bar{c} & \bar{c}.\bar{c} \end{vmatrix} = \left[\bar{a} \quad \bar{b} \quad \bar{c}\right]^3 = 125$

14. Let $f(1) = a, f(2) = b, f(3) = c$

Given $a + b + c = 1100$ and $S^3 = (a = 2b + 3c)^3 = 162$ abc

$\Rightarrow \dfrac{a + 2b + 3c}{3} = \sqrt[3]{6abc}$ Which is the case of $AM = GM$

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When $a=2b=3c \Rightarrow a=600, b=300, c=200$

Harmonic mean of $a,b\,c=\dfrac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}=\dfrac{3}{\frac{1}{600}+\frac{1}{300}+\frac{1}{200}}=300$

15. Arrange 600 red balls and there will be 599 gaps (excluding initial and final). So, minimum value will be $x_1=599-300-200=99$ For maximum arrange all 1100 and there will be 1099 gaps (excluding initial and final). So, maximum value will be $x_2=1101$ So, $x_1+x_2=1200$

16&17. $P_n=\dfrac{1}{3}P_{n-1}+\dfrac{2}{3}P_{n-1} \Rightarrow \dfrac{P_n-p_{n-1}}{P_{n-1}-P_{n-2}}=\dfrac{-1}{3}$

$\therefore P_2-P_1, P_3-P_2, P_4-P_3, \ldots\ldots$ are in G.P.

Also, $P_2-P_1=\dfrac{1}{9} \therefore P_n-P_{n-1}=\dfrac{1}{9}\left(\dfrac{-1}{3}\right)^{n-2}$

$\Rightarrow P_n-P_{n-1}=\dfrac{1}{9}\left(\dfrac{-1}{3}\right)^{n-2}$, $P_{n-1}-P_{n-2}=\dfrac{1}{9}\left(\dfrac{-1}{3}\right)^{n-3}$, $\ldots\ldots P_2-P_1=\dfrac{1}{9}$

$\Rightarrow P_n-P_1=\dfrac{1}{9}\left[\left(\dfrac{-1}{3}\right)^{n-2}+\left(\dfrac{-1}{3}\right)^{n-3}+\ldots.1\right]$

$\Rightarrow \lim\limits_{n\to\infty} P_n=P_1+\dfrac{1}{12}$

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PHYSICS

18. Consider a point P on side CD. Cut triangular section BCP and apply newtons laws. And maximise the force normal to side BP

19. Let's put the origin at a point O and point X the axis to the right and the axis $Y-$ down. Then the point O' - the centre of the small hoop - moves according to the law.

$$X(t)=r\sin\omega t,\ Y(t)=r\cos\omega t$$

if you count the time from the moment specified in the problem statement.
Let us denote through $\varphi(t)=\varphi_0-\omega_1 t$ the angle of rotation a small hoop rotating at rolling along a large hoop in the opposite direction with angular velocity ω_1. The law of motion of a point on a small hoop can be written as

$$x(t)=r\sin\omega t+r\sin\varphi(t),\ y(t)=r\cos\omega t+r\cos\varphi(t),$$

and the velocity and acceleration of this point are in the form of

$$\begin{aligned}v_x(t)&=r\omega\cos\omega t-r\omega_1\cos\varphi(t)\\ v_y(t)&=-r\omega\sin\omega t+r\omega_1\sin\varphi(t)\\ a_x(t)&=-\omega^2 r\sin\omega t-\omega_1^2 r\sin\varphi(t)\\ a_y(t)&=-\omega^2 r\cos\omega t-\omega_1^2 r\cos\varphi(t)\end{aligned}.$$

At the moment of time when $\varphi(t)=\omega t$, the point in question on the small hoop touches the large hoop, so its velocity should turn to zero. From this we get:

$$\omega_1=\omega,\ \text{и}\ a_x=-\omega^2 x,\ a_y=-\omega^2 y.$$

Therefore, at each moment of time, the acceleration of the point A on the small hoop is directed towards the center of the large hoop and is expressed in terms of the distance to the center of the large hoop as $\omega^2 R$. At the time shown in the figure in the problem condition, $R=r\sqrt{2}$. Thus, the acceleration vector of the point A is directed to the point O at an angle 45° to the segment AO' and is equal in magnitude to $a=\omega^2 r\sqrt{2}$.

20. If resistance of rheostat is increased current will decrease. Now since current decreases, so will the potential difference across each branch and hence voltmeter reading will increase.

21.

22. Let the gas do work during the cycle $A_0=\Delta p\Delta V$. Then $\eta_A=A_0/Q_{425}$, where the amount of heat supplied to the gas is $Q_{425}=U_{54}+A_{25}$. Similarly $\eta_B=A_0/Q_{798}$, where $Q_{798}=U_{87}+A_{98}$. Note that $A_{98}=A_{25}+A_0$. Compare the changes in internal energies U_{54} and U_{87}. The expression for U_{87} :

$$U_{87}=C_V(T_8-T_7)=\frac{C_V}{R}(p_3V_2-p_2V_1)=$$

$$=\frac{C_V}{R}(p_2V_2-p_1V_1+\Delta p\Delta V)=\frac{C_V}{R}(p_2V_2-p_1V_1+A_0).$$

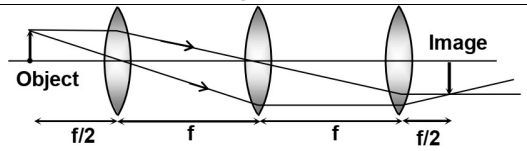
In a similar way we obtain that

$$U_{54}=\frac{C_V}{R}(p_2V_3-p_1V_2)=\frac{C_V}{R}(p_2V_2-p_1V_1+\Delta p\Delta V)=\frac{C_V}{R}(p_2V_2-p_1V_1+A_0),\ \text{that is}\ U_{54}=U_{87}.$$

From here $\eta_A=\frac{A_0}{Q_{425}}=\frac{A_0}{U_{54}+A_{25}}$, $\eta_B=\frac{A_0}{Q_{798}}=\frac{A_0}{U_{54}+A_{25}+A_0}=\frac{\eta_A}{1+\eta_A}$.

23. See figure for explanation

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24. Let L_1 and L_2, m_1 and m_2, V_1 and V_2 - be the length of the arms of the balance, the mass and volume of the balls, respectively.
The equilibrium condition on such scales: $m_1L_1 = m_2L_2$
Since the buoyant force is proportional to the volume of a given body, when immersed in water, the balance of the balance will not be disturbed if
$$V_1L_1 = V_2L_2$$
Divide term by term the first equality by the second: $\frac{m_1}{V_1} = \frac{m_2}{V_2}$ or $\rho_1 = \rho_2$
This suggests that the average densities of the balls are equal. From this it is clear that in a denser (silver) ball there is a cavity, as a result of which the equilibrium is maintained.

25. $m_Ac^2 + K_A + m_Bc^2 + K_B = m_Cc^2 + K_C +$ excitation energy So, $K_C = 2MeV$

26. For simplicity assume angular acceleration of plate to be zero. (Even if angular acceleration is considered answer will be same). From centre of mass frame acceleration of A, B C, and D will be equal in magnitude say a and toward O. Let the magnitude of acceleration of centre be b. The For magnitude of acceleration of A and C to be equal, b and a should be mutual perpendicular for A and C. $\sqrt{a^2+b^2} = 25$
And hence A and B will be in same direction. There fore $a+b=7$.
Hence magnitude of acceleration of D is $4-3=1m/s^2$

27. As the velocity V is given to lower mass, The tension in upper thread has to be zero or the thread has to be vertical so that the horizontal momentum of two mass system is conserved.



So in final state the upper thread is vertical and lower thread is just horizontal.
$2mV = mV_1 + 2mV_1$ Or final speeds of both balls is $V_1 = \frac{2V}{3}$
$\frac{2mv^2}{2} = \frac{mv_1^2}{2} + \frac{2mv_1^2}{2} + 2mgl$. Or $v = \sqrt{6gl}$.

28. Initial Energy of system is $\frac{1}{2}LI^2 + \frac{1}{2}LI^2 + MI^2 = 2LI^2$ because L=M
Initial Flux of each Loop is $LI + MI = 2LI$, Final Flux of each Loop is LI'
Therefore $I' = 2I$. Final Energy Of System is $\frac{1}{2}LI'^2 + \frac{1}{2}LI'^2 = 4LI^2$
Hence net wok done by external agent is $2LI^2$

29. We will consider the motion of our charged particle in a coordinate system moving along the X axis at a speed- v_0 i.e. towards the particle with speed v_0. In this coordinate system, an additional electric field appears with a strength equal to v_0B and directed towards the electric field $\vec{E}$. As a result, the electric field is zero, and only the magnetic field remains. A particle with an initial velocity $2v_0$, will move in a circle at a constant speed $2v_0$.

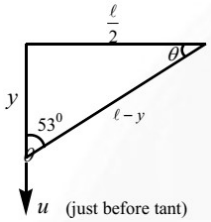
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Consequently, at the moment when the particle velocity vector will be 180° with vector $\vec{v}_0$, its speed will be equal to $2v_0$. If we now return to a fixed coordinate system, it is clear that the particle's speed will be equal to $3v_0$

30. MM-Collision Constraint. – Inelastic Collision, Constrained Motion $y=\frac{3\ell}{8}$ and $\theta=37^0$

The impulse on the bead will be at $\frac{53^0}{2}$ with the vertical.

Speed just after string is taut $v=\sqrt{2gy}\sin\left(\frac{53^0}{2}\right)$



31. The resistance of the plate between the vertices B and C before changesis ; $4r_2$

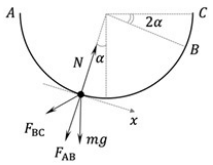
$$R_{AC}=\frac{\left(\frac{1}{2}Ir_1+2U_2\right)-\left(-\frac{1}{2}U_1-2Ir_2\right)}{I}=\frac{1}{2}\left(r_1+4r_2\right)+\frac{U_1+4U_2}{2I}$$

32. The resistance of the plate between the vertices A and C before changes is equal to

$$R_{AC}=\frac{1}{2}\left(r_1+4r_2\right)+\frac{U_1+4U_2}{2I}\,.\,R_{AC}=\frac{\left(\frac{1}{2}Ir_1+2U_2\right)-\left(-\frac{1}{2}U_1-2Ir_2\right)}{I}=\frac{1}{2}\left(r_1+4r_2\right)+\frac{U_1+4U_2}{2I}$$

33. $q\le\frac{\sqrt{18}mgR}{3k\lambda}$

34. (a) Move the bead from the equilibrium position to a small angle α from the vertical.



From symmetry, it follows that the section of AB the half-ring acts on the bead along the normal with the force F_{AB}, as shown in the figure. Let's write the condition for the existence of the returning force in the direction tangent to the semi-ring (axis Ox):

$mg\sin\alpha-F_{BC}\cos\frac{\pi}{4}\ge 0$ Since $F_{BC}=k\frac{q\lambda R\cdot 2\alpha}{2R^2}$ (directed at an angle $\frac{3\pi}{4}$ to the axis Ox), and the approximate formula is valid $\sin\alpha\approx\alpha$, we get $k\frac{q\lambda R\cdot 2\alpha}{2R^2}\le\frac{mg\alpha}{\cos\frac{\pi}{4}}\Rightarrow q\le\frac{\sqrt{2}mgR}{k\lambda}$

b) Let's write down Newton's second law for a bead (see figure):

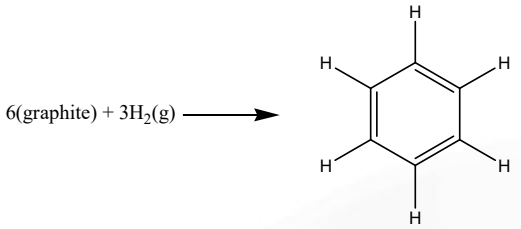
$$-m\ddot{x}=mg\sin\alpha-F_{BC}\cos\frac{\pi}{4}\approx mg\frac{x}{R}-k\frac{q\lambda\cdot x}{R^2}\cdot\frac{\sqrt{2}}{2}.$$

Therefore, the period of small oscillations is equal to $T=2\pi\sqrt{\frac{\sqrt{2}mR^2}{\sqrt{2}mgR-kq\lambda}}$.

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CHEMISTRY

35.

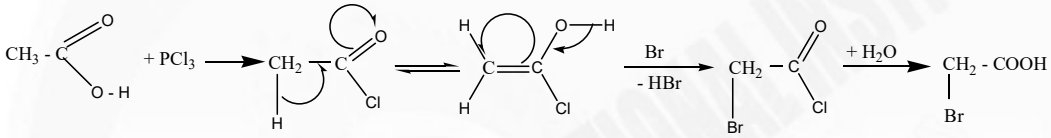


When we assume that there is no resonance in the benzene, in benzene $3C-C, 3C=C, 6C-H$ bonds present theoretical heat of formation of $C_6H_6(g)$ by assuring there is no resonance $=6\times718+3\times436-6\times416-3\times331-3\times591$

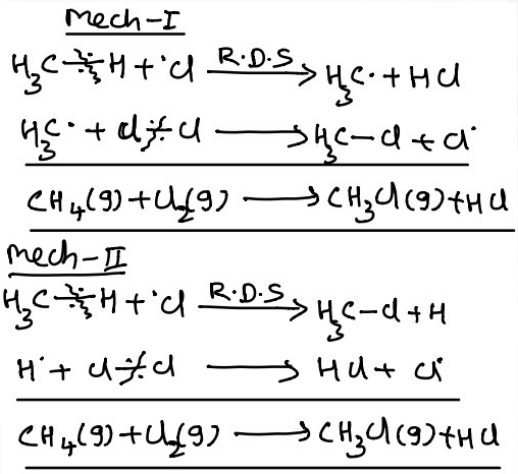
$=354\,KJ/mol$

Resonance energy of $C_6H_6(g)=83-354=-271KJ/mol$

36. α – bromination, but not α – chlorination



37.

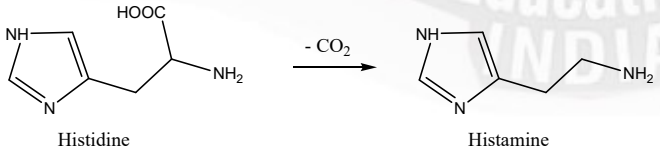


Free radical attack is highly influenced by steric factor. Free radical preferentially attack the monovalent atom (i.e less crowded atom) than the polyvalent atom (i.e more crowded atom). Mech-I Correct, Mech-II is incorrect.

38. Plane which contains maximum number of atom is 3.

39.

40.



b) From 1 mol histamine 1.5 mol N_2 gas formed.

d) brompherinamine is antagonist but not against

41. Tetrahedral complex is $\left[NiCl_2\left(P(C_6H_5)_3\right)_2\right]$

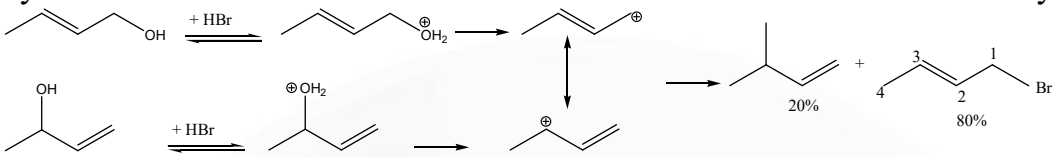
42. Number of possible Aldohexopyranoses are $2^5=32$

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Number of possible $D-$aldohexopyranoses are $2^5/2=2^4=16$

Number of possible $d-D-$Aldohexopyranoses are $2^4/2=2^3=8$

43. In reaction of unsymmetrical allyl alcohols with HBr, the regioselectricity is determined by steric hindrance. Attack of Br^- is factor at the less hindered end of allylic system.



44. $P=\sqrt{P_A^e.P_B^e}=\sqrt{100\times900}=300\text{ torr}$

45. $2\to1$ in hydrogen and $4\to2$ in He^+ have same wavelength

$\therefore$ No. of spectral lines = 10+10-1=19

46. Hydrated oxide are given by Iron and Aluminium

47. Trigonal bipyramidal structure and planar atoms are 8

48. $\dfrac{h}{1-h}=\sqrt{K_n}$

$\dfrac{h}{1-h}=\sqrt{\dfrac{K_w}{K_a.K_b}}$

$\dfrac{h}{1-h}=\sqrt{\dfrac{10^{-14}}{3\times10^{-7}\times3\times10^{-7}}}\Rightarrow\dfrac{h}{1-h}=\dfrac{1}{3}\Rightarrow3h=1-h\Rightarrow h=0.25$

% of hydrolysis = 25%

49. $HA+BOH\rightleftharpoons BA+H_2O$

$K_c=\dfrac{K_a.K_b}{K_w}=\dfrac{3\times10^{-7}\times3\times10^{-7}}{10^{-14}}=9=9\times10^{-0}$

50. $Mg_3(OH)_2[Si_6O_{10}]$

51. $[Be_2Al_2Si_6O_{18}]$

Unshared oxygens are 12

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